VISHNU R NAIR

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PROFILE

CSE (AI & ML) undergraduate with Python and C foundations, developing practical understanding of APIs, relational data and information retrieval through project-based learning. Seeking an internship with hands-on engineering work and mentorship.

EDUCATION

Sree Chitra Thirunal College of Engineering (SCTCE), Thiruvananthapuram
B.Tech, Computer Science and Engineering (AI & ML) | Expected graduation: 2029
APJ Abdul Kalam Technological University (KTU)

TECHNICAL FOUNDATIONS & LEARNING

Current foundations: Python, C
Learning through the projects below: FastAPI, SQLAlchemy, SQL, HTML, CSS, JavaScript, scikit-learn, Git and automated testing.

PROJECT PORTFOLIO

CampusTrack | Python, FastAPI, SQLAlchemy, JavaScript

- Implemented an application tracker with account registration, saved records, stage filtering, deadlines and CSV export; SQLite locally with PostgreSQL deployment configuration.
- Added salted password hashing, expiring sessions, ownership checks and CSRF protection; tests cover cross-account isolation, persistence and unsafe input.

NoteLens | Python, scikit-learn, NumPy, FastAPI

- Implemented study-note retrieval using TF-IDF, truncated SVD and hybrid ranking, returning source filenames, page numbers and cited passages from text and PDF notes.
- Added isolated temporary workspaces, bounded PDF processing and a reproducible small ranking comparison; documented dataset limits and similarity-score interpretation.

RepoCheck | Python, FastAPI, CLI

- Implemented a shared rule engine for a local CLI and API to inspect repository hygiene and selected risky code patterns without executing source.
- Added severity filters, redacted JSON reports and archive limits; tests cover unsafe paths, excessive compression and omission of matched values.

Personal Portfolio | HTML, CSS, JavaScript

- Created a responsive site with semantic page structure, real application screenshots, keyboard-accessible project dialogs and a prepared GitHub Pages workflow.

AI-assisted learning builds with published source repositories. Python projects passed GitHub Actions; live deployments and independent code walkthroughs are in progress.